PORTFOLIO PROJECT · HEALTHCARE CLAIMS AUTOMATION

FWA Claims Rule Engine

A rule engine that screens health-insurance claims for fraud, waste and abuse. 32 rules check every claim and service line, explain each finding, and work with any claims export layout.

Built on Saudi standards (NPHIES SBS, ICD-10, SFDA, GTIN); the code lists can be swapped for CPT, HCPCS, ICD-10-CM or NDC.

32 validation rules

Run over every claim and service line

5 claim-level rules

Judge whole claims: duplicates, clones, capacity

27 service-line rules

Judge each line: coding, frequency, pharmacy

Any layout, any size

Export columns mapped once and remembered

Open the rule flowcharts →

The problem

Claim audit is mostly manual: reviewers sample claims and hunt for the same patterns, such as duplicate services, diagnoses that don't fit the patient's age or sex, cloned claims, duplicate drugs, or visits after a policy expired. Each check is simple; running all of them on every line is not.

This engine turns those checks into documented rules that run on the whole file in one pass.

How it works

STEP 1

Map the export

Headers are matched once to a 60-column template and remembered.

STEP 2

Load the standards

SBS, ICD-10, SFDA and GTIN lists are stored and reused.

STEP 3

Run the rules

Unpaid lines are set aside; claim and service-line rules run in a fixed order.

STEP 4

Deliver the findings

An Excel workbook with a reason next to every flagged line.

STEP 5

Track it on the dashboard

Findings turned into money and trends by rule and provider.

Why the dashboard matters

The findings list says what's wrong; the dashboard says where to act first.

- **Headline cards**: paid vs rejected amount and the rejection rate, each line counted once.
- **Trends**: billed and rejected amounts per provider over time.
- **By rule**: which rules carry the rejected money.
- **Provider behaviour**: unlisted-code shares and over-used services.
- **Filters**: provider, period and service type.

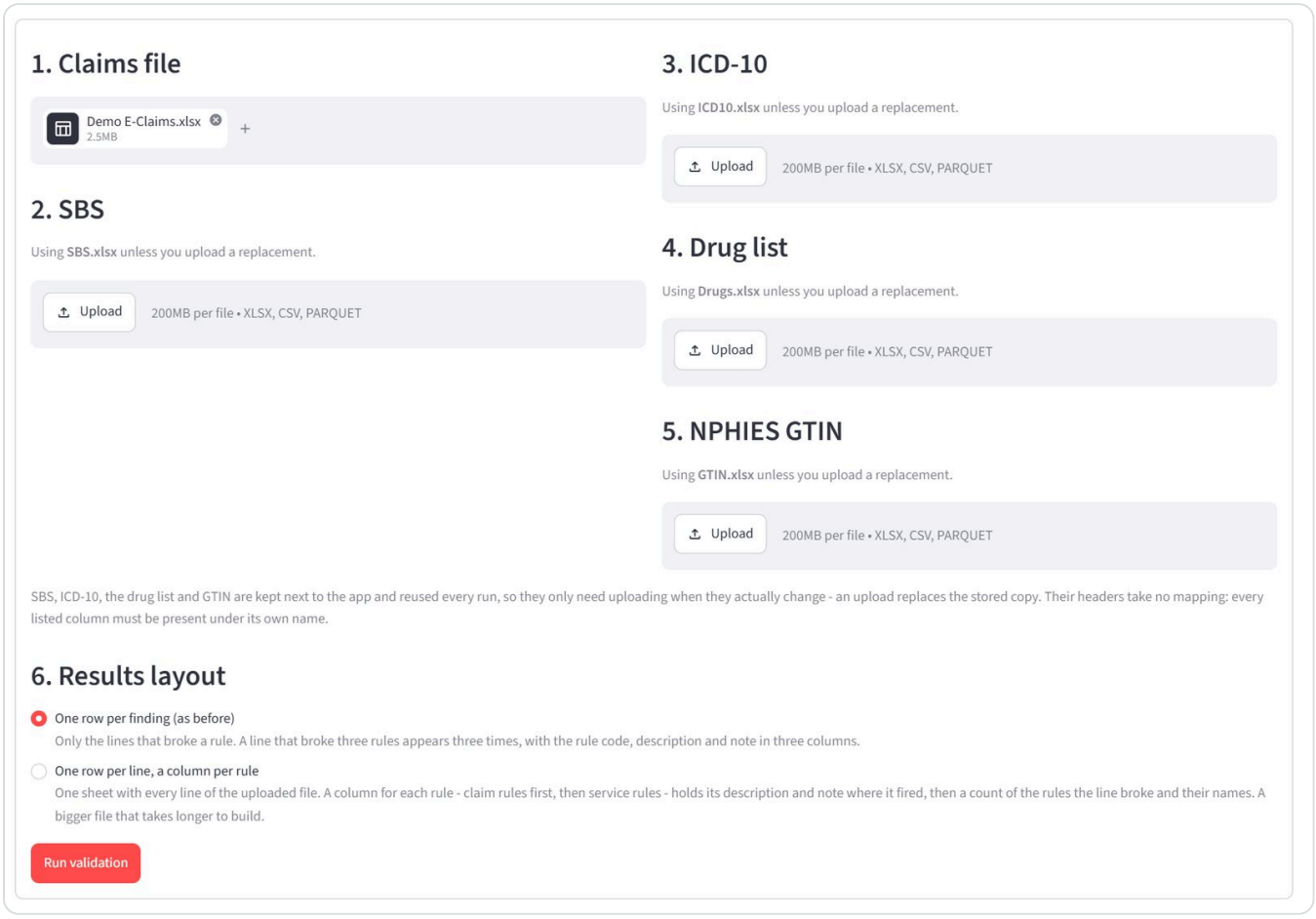
See it in action

> Screens from the app, run on a synthetic claims file. All patients, physicians, providers and claims are invented.

> ▶ Demo: the app running end to end, one minute: watch it in the online version, https://claude.ai/artifact/FRE893ZRpnVcErYrCKHSpv

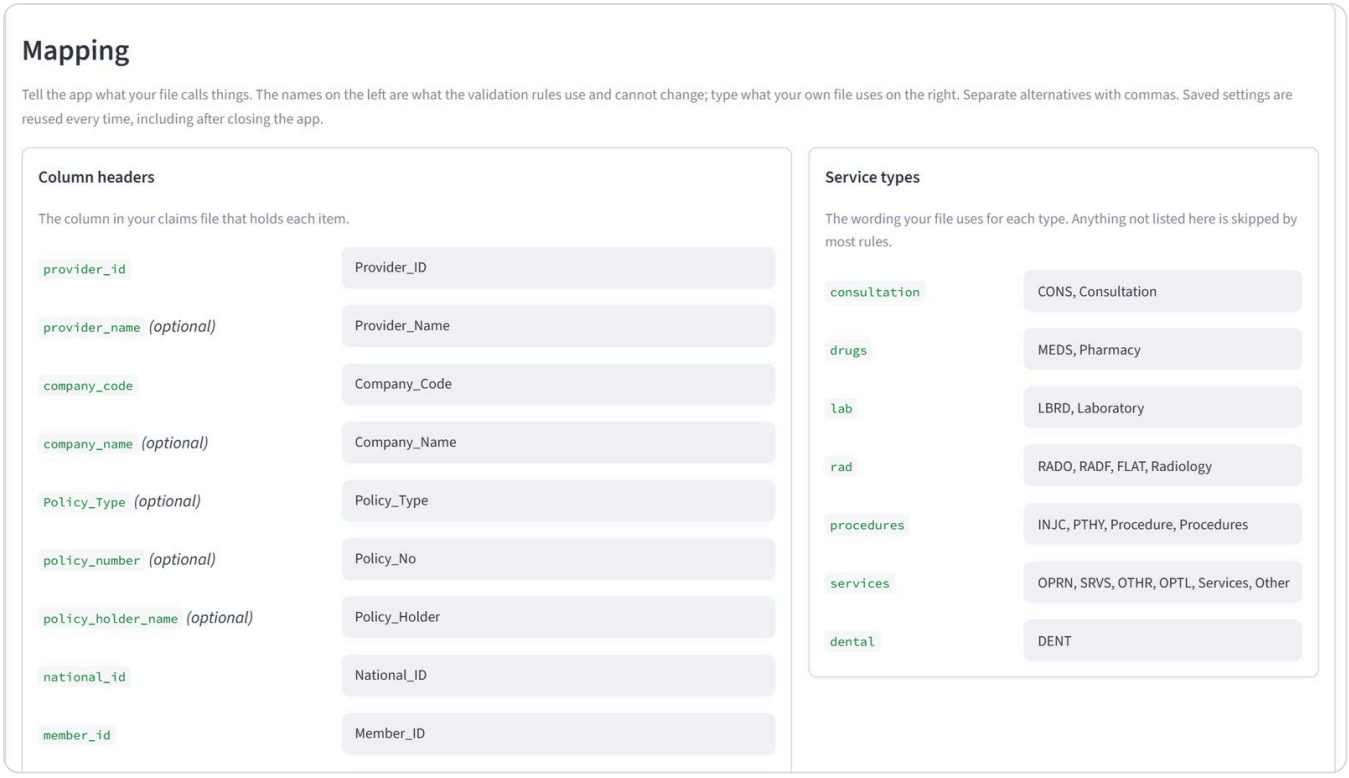
Run screen

Upload the claims file and choose the results layout.



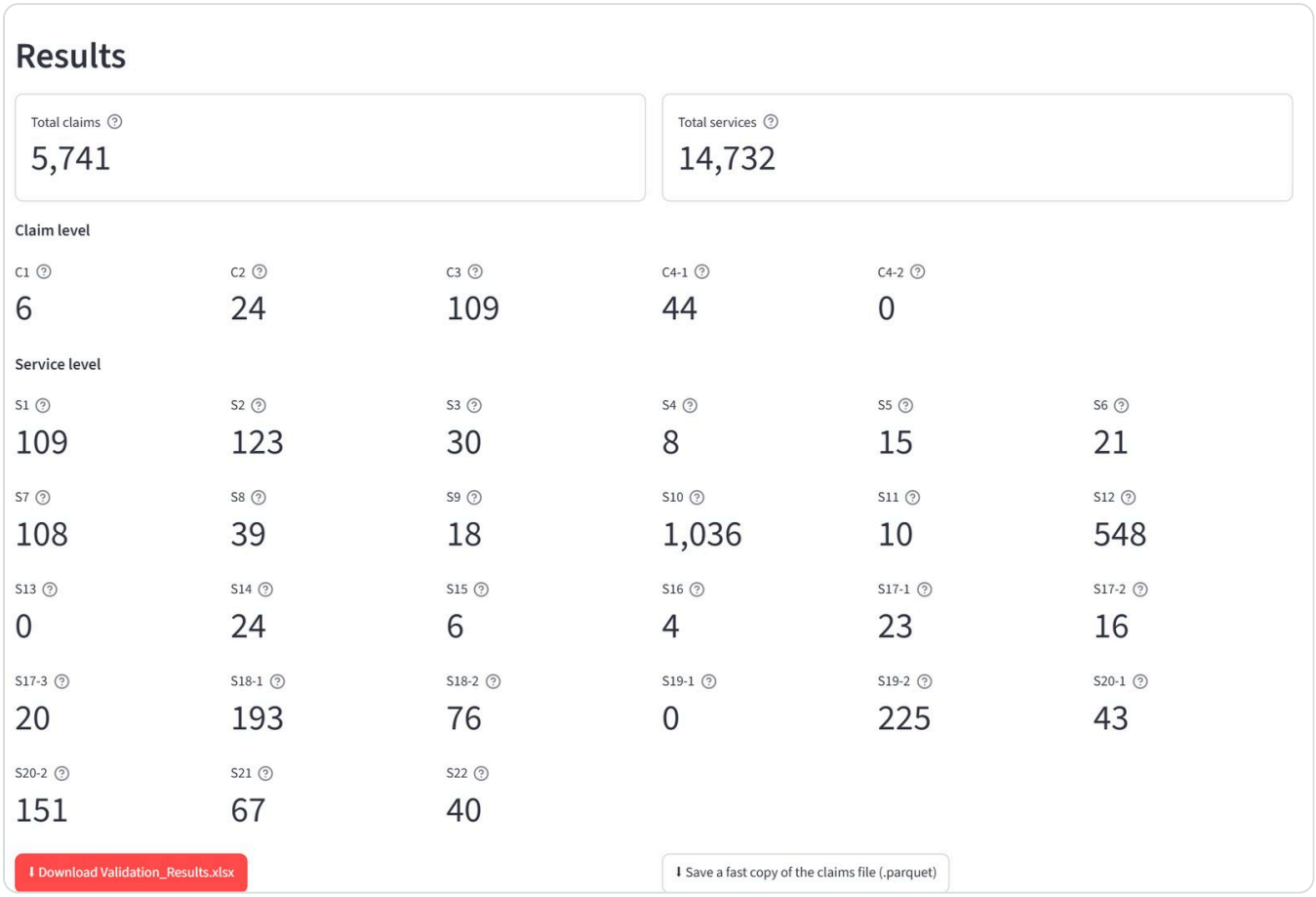
Column converter

Map the file's headers and service types once; they're reused every run.



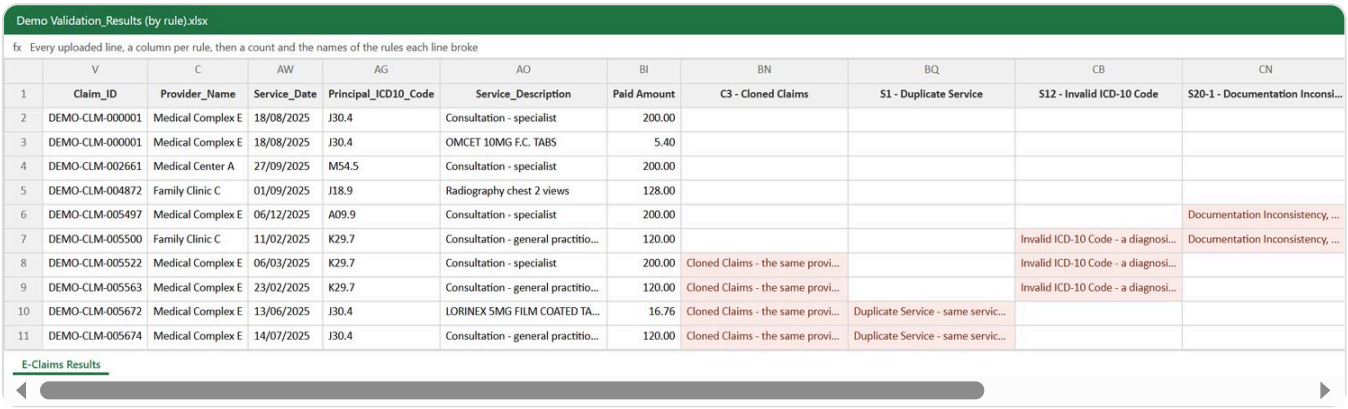
Results

Total claims and service lines, then a count per rule.



What the reviewer gets

Two layouts: **one row per finding**, or **one row per line** with a column for every rule plus a count and the names of the rules each line broke. Below, a slice of the demo workbook (14,732 lines, 97 columns).



Demo Validation_Results (by rule).xlsx

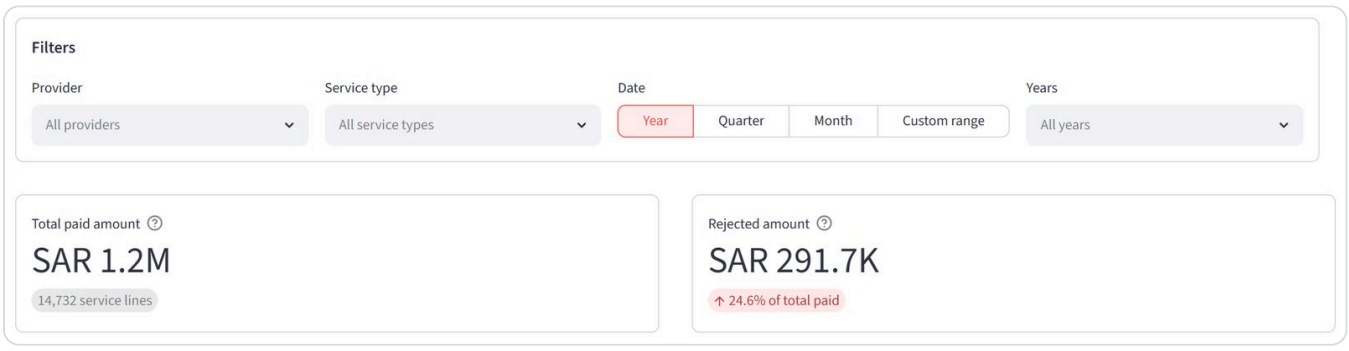
fx Every uploaded line, a column per rule, then a count and the names of the rules each line broke

	V	C	AW	AG	AO	BI	BN	BQ	CB	CN
1	Claim_ID	Provider_Name	Service_Date	Principal_ICD10_Code	Service_Description	Paid Amount	C3 - Cloned Claims	S1 - Duplicate Service	S12 - Invalid ICD-10 Code	S20-1 - Documentation Inconsi...
2	DEMO-CLM-000001	Medical Complex E	18/08/2025	J30.4	Consultation - specialist	200.00				
3	DEMO-CLM-000001	Medical Complex E	18/08/2025	J30.4	OMCET 10MG F.C. TABS	5.40				
4	DEMO-CLM-002661	Medical Center A	27/09/2025	M54.5	Consultation - specialist	200.00				
5	DEMO-CLM-004872	Family Clinic C	01/09/2025	J18.9	Radiography chest 2 views	128.00				
6	DEMO-CLM-005497	Medical Complex E	06/12/2025	A09.9	Consultation - specialist	200.00				Documentation Inconsistency, ...
7	DEMO-CLM-005500	Family Clinic C	11/02/2025	K29.7	Consultation - general practitio...	120.00			Invalid ICD-10 Code - a diagnosi...	Documentation Inconsistency, ...
8	DEMO-CLM-005522	Medical Complex E	06/03/2025	K29.7	Consultation - specialist	200.00	Cloned Claims - the same provi...		Invalid ICD-10 Code - a diagnosi...	
9	DEMO-CLM-005563	Medical Complex E	23/02/2025	K29.7	Consultation - general practitio...	120.00	Cloned Claims - the same provi...		Invalid ICD-10 Code - a diagnosi...	
10	DEMO-CLM-005672	Medical Complex E	13/06/2025	J30.4	LORINEX 5MG FILM COATED TA...	16.76	Cloned Claims - the same provi...	Duplicate Service - same servic...		
11	DEMO-CLM-005674	Medical Complex E	14/07/2025	J30.4	Consultation - general practitio...	120.00	Cloned Claims - the same provi...	Duplicate Service - same servic...		

E-Claims Results

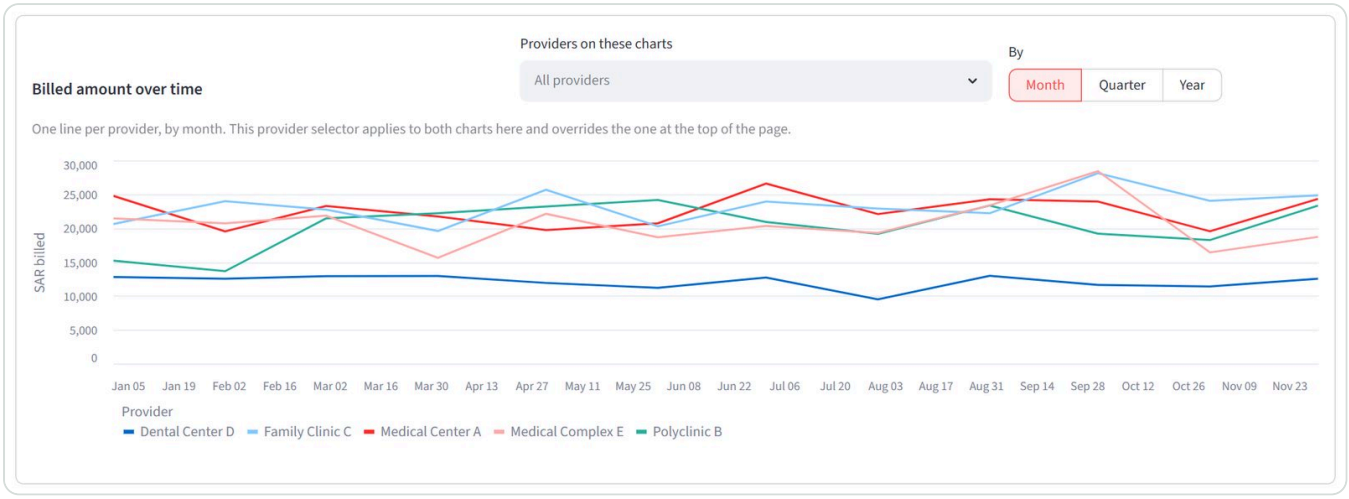
Dashboard: overview

Filters and paid vs rejected totals.



Dashboard: trend by provider

Billed amount per provider over time.



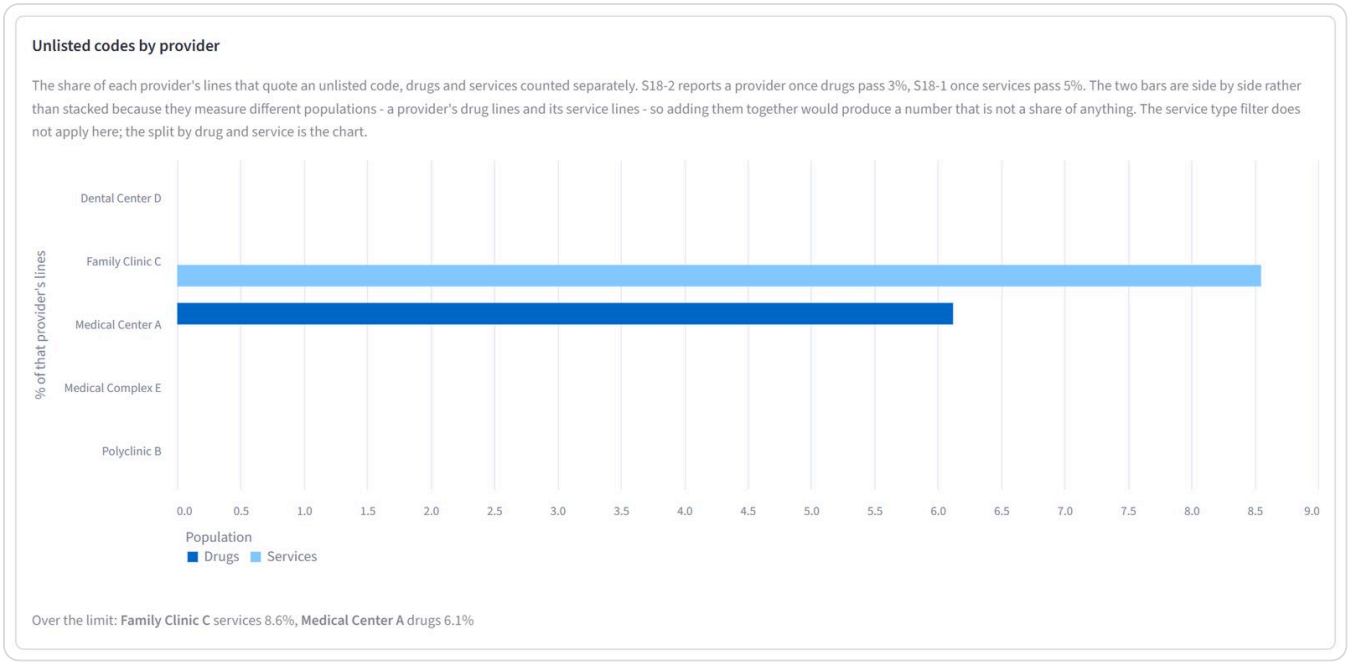
Dashboard: where the money is

Rejected amount by rule.



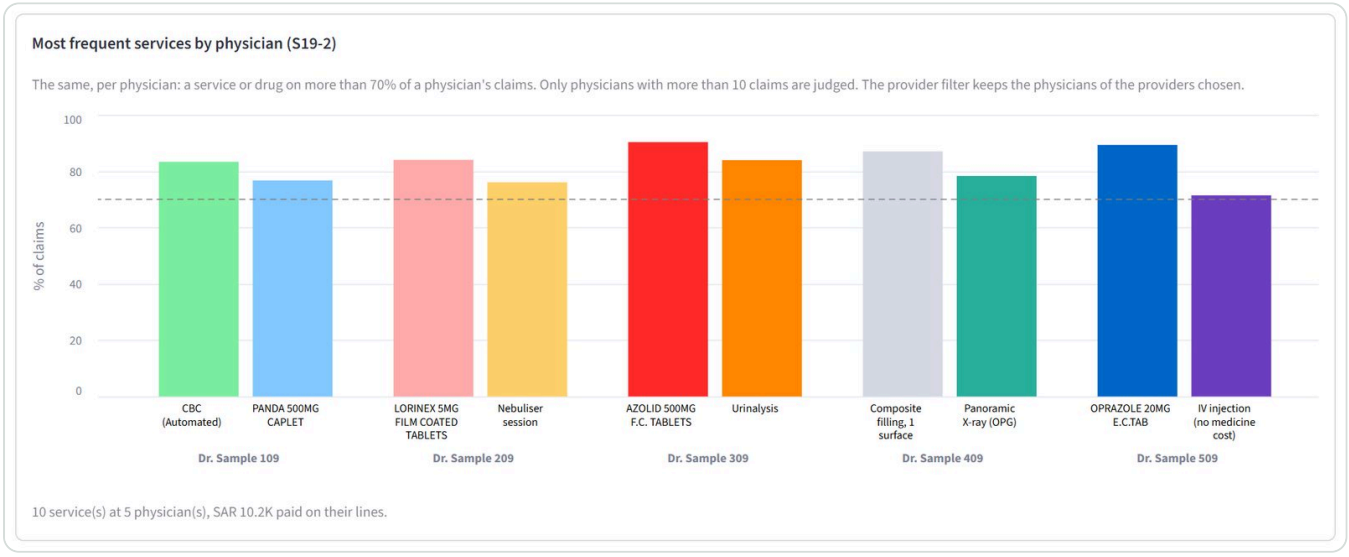
Dashboard: provider behaviour

Each provider's unlisted-code share against the S18 limits.



Dashboard: physician pattern

Services each physician attaches to over 70% of their claims (rule S19-2).



The 32 rules

Each rule has a flowchart and a note template, so every finding says why it was raised.

Open the rule flowcharts →

Claim level `5 rules`

C1	**Duplicate Claim** Two or more claims for the same patient on the same day billing exactly the same set of services.
C2	**Member Treated at Multiple Providers, Same Day** A patient with paid consultation lines at two or more different providers on one day.
C3	**Cloned Claims** The same provider, anywhere in the file, billing an identical set of services, principal diagnoses and chief complaint for different patients.
C4-1	**Physician Daily Capacity Breach (doubtful)** A physician seeing more than 40 different patients in consultation on one day.
C4-2	**Physician Daily Capacity Breach (confirmed)** A physician seeing 60 or more different patients in consultation on one day.

Duplication and frequency `9 rules`

S1	**Duplicate Service** The same service billed more than once for the same patient on the same day.
S2	**Repeat Consultation Within 14 Days** The same patient seen again by the same physician 1 to 14 days after the previous visit.
S3	**Service Frequency Breach** A service repeated sooner than the frequency SBS sets for its NPHIES code, counted across providers.
S4	**Excessive Radiology** More than 3 radiology studies for the same patient in a rolling 14 days.
S5	**Exceeds Maximum Daily Limit** A line billed with a quantity greater than the daily limit SBS sets for its NPHIES code.
S6	**Unreasonable Quantity** A drug line billed with a quantity above 5.

S7

Duplicated Service Within 7 Days

The same service for the same patient at the same provider repeated 1 to 7 days later.

S11

Excessive Consultation Frequency

A patient with more than 3 consultation-only claims inside a rolling 30 days.

S22

Excessive Member Visits

A patient with more than 5 visit days inside a rolling 30 days, counted across every provider and service type.

Coding validity `7 rules`

S8

Diagnosis Outside Age Range

A diagnosis code billed for a patient whose age at the service date is outside the range ICD-10 allows.

S9

Diagnosis Gender Mismatch

A diagnosis code ICD-10 strictly restricts to one sex, billed for a patient of the other.

S12

Invalid ICD-10 Code

A diagnosis code marked valid = 0 in ICD-10, or absent from the reference altogether.

S13

Inactive ICD-10 Code

A diagnosis code ICD-10 had withdrawn by the service date and not reactivated.

S14

Not Valid Principal Diagnosis

A primary diagnosis that ICD-10 allows only as an additional diagnosis.

S15

Service / Gender Mismatch

A service SBS restricts to one sex, billed for a patient of the other.

S16

Service / Age Mismatch

A service billed for a patient outside the age range SBS allows it for.

Pharmacy `3 rules`

S17-1

Therapeutic Duplication, Same Active Ingredient

Two different drugs given to one patient on one day that share the ATC code to 7 characters (same active ingredient).

S17-2

Therapeutic Duplication, Same Chemical Subgroup

Two different drugs given to one patient on one day that share the ATC code to 5 characters (same chemical subgroup).

S17-3

Therapeutic Duplication, Same Pharmacological Subgroup

Two different drugs given to one patient on one day that share the ATC code to 4 characters (same pharmacological subgroup).

Provider and physician behaviour `4 rules`

S18-1

More Than 5% Unlisted SBS Codes

Every unlisted service from a provider whose unlisted share of non-drug lines is 5% or more.

S18-2

More Than 3% Unlisted Drugs

Every unlisted drug from a provider whose unlisted share of drug lines is 3% or more.

S19-1

Service Too Frequent Use, Provider Level

A service or drug that appears on more than 70% of a provider's claims, consultation excluded.

S19-2

Service Too Frequent Use, Physician Level

A service or drug that appears on more than 70% of a physician's claims, consultation excluded.

Documentation and coverage `4 rules`

S10

Isolated Consultation Visit

A claim billing nothing but consultation, with no test, medicine or procedure that followed.

S20-1

Phantom Billing

A paid line on a claim whose chief complaint says the visit was cancelled or the patient not seen.

S20-2

Missed Chief Complaint

A paid line on a claim with no chief complaint, or one holding no real word.

S21

Service Outside Coverage

A service dated before the policy start date or after the policy end date.

Built to fit any claims data

Any layout

- A 60-column template, mapped once to the file's own headers.
- Missing fields are created blank; rules that need them say so.
- Results keep the file's own header names.

Any size

- Rules work on whole columns, so there's no line limit.
- Results past Excel's row limit continue on a second sheet.
- Repeat runs load in seconds from a cached copy.

How the rules were set and proven

📏 Thresholds from the data

Limits were measured, not guessed; the excessive-visits limit, for example, came from Tukey's outlier fences.

🎯 Tested at the boundary

Each rule is tested at its exact limit: quiet at the threshold, firing one past it.

🔁 Checked against a recount

Every new or changed rule is recounted independently, line by line, and the other rules confirmed unchanged.

📝 Documented for review

A reference workbook, a flowchart per rule, and a review page for stakeholder comments.

My role and tools

I designed the rules, set and tested the thresholds, defined how findings are reported, and validated the results. Built in Python (pandas) with a Streamlit app, using AI-assisted development.

Python · pandas · Streamlit · Altair · Excel · NPHIES SBS · ICD-10 · SFDA drug register · GTIN

Personal portfolio project. No real claims, patient or provider data is shown.